

PRIME PALINDROME PROGRAM

1. AIM

To write a Java program to check whether a given number is both prime and palindrome.

2. DESCRIPTION

The program reads an integer from the user and uses separate methods to check whether the number is prime and whether it reads the same forwards and backwards. It then displays the appropriate result.

3. SYSTEM REQUIREMENTS

- Java Development Kit (JDK)
- Java compiler and Java Runtime Environment (JRE)
- Any Java IDE such as Eclipse, IntelliJ IDEA, or NetBeans
- Operating system: Windows, Linux, or macOS
- Minimum 2 GB RAM

4. ALGORITHM

1. Start the program.
2. Read the number from the user.
3. Check whether the number is prime.
4. Reverse the number and check whether it is a palindrome.
5. If both conditions are true, display PRIME AND PALINDROME.
6. Otherwise display the corresponding result.
7. Stop the program.

5. UML CLASS DIAGRAM

PrimeCheck	PrimePalindrome
-----	-----
+ isPrime(n:int): boolean	+ main(args:String[]): void
+ isPalindrome(n:int): boolean	-----

6. COMPLETE PROGRAM CODE

```
package palindromeprog;

import java.util.Scanner;

class PrimeCheck {

    static boolean isPrime(int n) {
        if (n <= 1)
            return false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0)
                return false;
        }
    }
}
```

```

        return true;
    }

    static boolean isPalindrome(int n) {
        int original = n;
        int reverse = 0;

        while (n > 0) {
            int digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }
        return original == reverse;
    }
}

public class PrimePalindrome {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");
        int number = sc.nextInt();

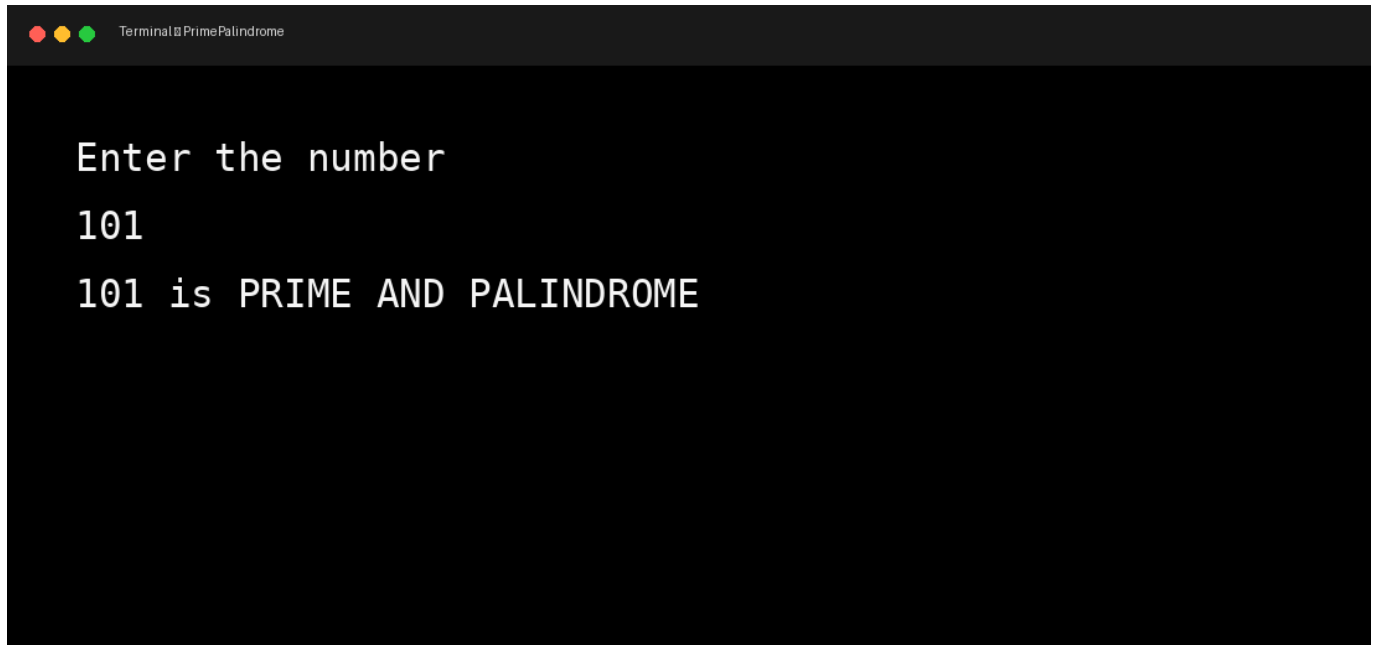
        boolean prime = PrimeCheck.isPrime(number);
        boolean palindrome = PrimeCheck.isPalindrome(number);

        if (prime && palindrome) {
            System.out.println(number +
                " is PRIME AND PALINDROME");
        } else if (prime) {
            System.out.println(number +
                " is PRIME BUT NOT PALINDROME");
        } else if (palindrome) {
            System.out.println(number +
                " is NOT PRIME BUT PALINDROME");
        } else {
            System.out.println(number +
                " is NEITHER PRIME NOR PALINDROME");
        }

        sc.close();
    }
}

```

7. OUTPUT SCREENSHOT

A screenshot of a terminal window with a dark background. The title bar at the top shows three colored circles (red, yellow, green) and the text "Terminal @ PrimePalindrome". The terminal content shows the prompt "Enter the number", the input "101", and the output "101 is PRIME AND PALINDROME".

```
Terminal @ PrimePalindrome

Enter the number
101
101 is PRIME AND PALINDROME
```

8. RESULT

Thus, the Java program to check whether a number is prime and palindrome was successfully implemented and executed.